

Parity-Last-4: one algorithmic task, four SR subproblems

$$y_t = (x_t + x_{t-1} + x_{t-2} + x_{t-3}) \bmod 2$$

Return 1 when the last four bits contain an odd number of ones.

Input	1	0	1	1	0	0	1	0
Output	1	1	0	1	0	0	0	1

Missing preceding bits are zero; the final four inputs are highlighted.

Refine the learned integer encoding

round(Z)	→	round(4Z)
12 observed states		16 observed states
Conflicting transitions		All relations deterministic

Three state updates + one output function

Exact recovery: successful seeds / 10

Scalar relation	Input-table size	Base PySR	SRBench-evolved 709715	MIPS-evolved 709714
$h_{t,0} = f_0(h_{t-1}, x_t)$	32 state-input pairs	0/10	0/10	1/10
$h_{t,1} = f_1(h_{t-1}, x_t)$	32 state-input pairs	5/10	3/10	10/10
$h_{t,2} = f_2(h_{t-1}, x_t)$	32 state-input pairs	10/10	10/10	10/10
$y_t = g(h_t)$	16 states	10/10	10/10	10/10
Subproblems recovered across seeds		3/4	3/4	4/4